Vigneshwaran Radhakrishnan

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Professional Preparation

May 2024 Postdoctoral Researcher in Aerospace Engineering
Current *Texas A&M University, College Station, Texas*

Aug 2017 Ph.D. in Aerospace Engineering
Apr 2024 *Texas A&M University, College Station, Texas*

Advisor: Amine Benzerga
Dissertation: Three-Dimensional Simulations of Ductile Fracture under Arbitrary Loadings

Aug 2014 Master of Technology in Aerospace Engineering
May 2016 *Indian Institute of Technology-Kanpur, India*

Advisors: P.M.Mohite, C.S.Upadhyay
Thesis: Numerical Assessment of GLD Plasticity Model for Porous Materials

Aug 2010 Bachelor of Engineering in Aeronautical Engineering
May 2014 *Madras Institute of Technology, Anna University, India*

Advisors: R. Dhanaraj, H.S.N. Murthy
Project: Optimisation of Ultrasonic Transducer Configuration for Detection of Cracks

Professional Experience

Mar 2024	-	Apr 2024	Research Engineer	Aero. Engg., Texas A&M University
Feb 2017	-	Aug 2024	Research Assistant	Aero. Engg., Texas A&M University
Aug 2018	-	Dec 2018	Teaching Assistant	Aero. Engg., Texas A&M University
Jul 2016	-	Jul 2017	Engineer	VEM Technologies Pvt. Ltd., Hyderabad, India
Jul 2015	-	Dec 2015	Teaching Assistant	Aero. Engg., Indian Institute of Technology-Kanpur

Publications

[1] **Vigneshwaran, R.** and A A Benzerga. “Unhomogeneous yielding of porous materials — Evolution Equations”. In: *Journal of the Mechanics and Physics of Solids* 196 (2025), p. 105973.

[2] **Vigneshwaran, R.** and A. A. Benzerga. “Assessment of iterated variational homogenization estimates for microstructure evolution under unhomogeneous yielding”. In: *SSRN* (2025). (submitted for peer review).

[3] **Vigneshwaran, R.** and A. A. Benzerga. “An analysis of failure in shear versus tension”. In: *European Journal of Mechanics-A/Solids* 104 (2024), p. 105074.

[4] **Vigneshwaran, R.** and A. A. Benzerga. “Criterion for unhomogeneous yielding of porous materials”. In: *Journal of the Mechanics and Physics of Solids* 192 (2024), p. 105804.

[5] **Vigneshwaran, R.** and A. A. Benzerga. "Assessment of a two-surface plasticity model for hexagonal materials". In: *Journal of Magnesium and Alloys* 11.12 (2023), pp. 4431–4444.

Conference Proceedings:

[6] **Vigneshwaran, R.** and Amine Benzerga. "Finite Element Analysis of Ductile Failure Under Severe Shear". In: *AIAA Scitech 2025 Forum*. 2025, p. 1360.

[7] **Vigneshwaran, R.** and A. A. Benzerga. "A Predictive Multisurface Approach to Damage Modeling in Mg Alloys". In: *Magnesium Technology 2022*. Cham: Springer International Publishing, 2022, pp. 293–297.

In Preparation:

[8] S. Datta, **Vigneshwaran, R.**, S. P. Joshi, and A. A. Benzerga. *Assessment of reduced order models for Magnesium alloys*.

[9] **Vigneshwaran, R.** and A. A. Benzerga. *Unhomogeneous yielding of porous materials — Simple criterion*. (to be submitted).

Invited talks

[1] **Vigneshwaran, R.** and A. A. Benzerga. "Ductile failure of a top-hat shear specimen". Mechanics of materials seminar, Lawrence Livermore National Laboratories. 2025.

[2] **Vigneshwaran, R.** and A. A. Benzerga. "A Predictive Multisurface Approach to Damage Modeling in HCP Alloys". Engineering and Applied Science Forum for Graduate Students, Postdoctoral Fellows, and Young Faculties. 2022.

[3] **Vigneshwaran, R.**, S. Wajid, D. Mortari, and A. A. Benzerga. "On limit analysis using functional interpolation". Theory and Applications of Functional Interpolation to Optimization and Control, U.S. National Congress on Theoretical and Applied Mechanics. 2022.

Honors and Awards

- **SES 2025- Future Faculty Symposium Award**
  Awarded for participation at the 2nd future faculty symposium at the 2025 SES Annual Technical Conference.

- **Postdoctoral Mentoring Fellow**
  A certification by the postdoctoral association, Texas A&M University.

- **ASME: IMECE 2022 - Honorable Mention**
  Awarded at the 2022 CONCAM Student Poster Competition on Computational Mechanics by the ASME during the International Mechanical Engineering Congress & Exposition in Columbus, OH, USA.

- **USNC: 17th Travel Grant Award**
  Awarded complimentary registration to the 17th U.S. National Congress on Computational Mechanics.

- **ICF: 15th Travel Grant Award**
  Awarded complimentary registration to the 15th International Conference on Fracture.

- **SES 2022 Conference Funding Award**
  Awarded complimentary registration to the 2022 SES Annual Technical Conference as a Student Volunteer Attendee.
- **Aero Graduate Excellence Fellowship Award**
  Awarded for Academic Excellence in graduate school at Texas A&M University.
- **Certificate of Merit**
  Awarded for Academic Excellence in the Master of Technology in Aerospace Engineering at IIT Kanpur.
- **Gold Medalist**
  Achieved in B.E. Aeronautical Engineering at Madras Institute of Technology, Anna University.

Teaching and Mentoring Experience

Aug 2024 – Dec 2024 — Substitute Instructor, Aerospace Engineering, Texas A&M University
AERO304 - Aerospace Structural Analysis, Instructor: Amine Benzerga

Delivered lectures on concepts of strain, traction, stress, and beam bending to a class of 90 students over three weeks.

Aug 2018 – Dec 2018 — Graduate Assistant, Aerospace Engineering, Texas A&M University
AERO306 - Aerospace Structural Analysis, Instructor: John Whitcomb

Reviewed and graded quizzes, assignments, two mid-term exams, and a final exam for 40 students. Prepared and delivered two hours of lectures on virtual work methods in structural analysis.

Jul 2015 – Dec 2015 — Teaching Assistant, Aerospace Engineering, Indian Institute of Technology-Kanpur
AE451 - Experiments in Aerospace Engineering, Instructor: Rajesh Kitey

Taught concepts related to the experiment on 'Combined Loading of Aluminum Alloy'. Graded laboratory reports for 50 students.

Jan 2019 – July 2021 — Mentor for Undergraduate Student, Aerospace Engineering, Texas A&M University
Advisor: Amine Benzerga

UG Student: Osvaldo Quintero
Trained Osvaldo to use ABAQUS software and guided him to develop a prototype ductile failure code that generates data for a machine learning model.

Course works

Aerospace Structural Analysis
Composite Materials
Continuum Mechanics
Failure Mechanics
Finite Element Methods
Fluid Mechanics
Functional Interpolation
Micromechanics
Multiscale Mechanics
Nonlinear FEM
Nonlinear Elasticity
Numerical Methods of PDE
Plasticity Theory
Rocket and Missile Structures
Smart Structures
Spacecraft Dynamics

Service

- Reviewer of the Journal of Mechanics Research Communications, and Scientific Reports.
- Undergraduate thesis proposal reviewer at Texas A&M University.
- Judge at the Student Research Week of 2025, 2022 and 2021 at Texas A&M University: Judged sessions, ranked, and nominated presentations for awards at the largest student-run research symposium.
- Member of Texas Running Against Cancer (TRAC).

Professional Affiliations

- American Society of Mechanical Engineers (ASME)
- American Institute of Aeronautics and Astronautics (AIAA)
- ASM International
- Aircraft Owners and Pilots Association (AOPA)